End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name: B.Tech (CSE, ECE), Semester: First
(March, 2023)

ubject Code: BAS-103	Subject: Applied Physics-I
ime :3 Hours	Maximum Marks :60

Note: Q. 1 is compulsory.
Attempt one question each from the Units I, II, III & IV.

Q1 (5*4=20)

(a) Unpolarized light is incident on two polarizing sheets placed one on top of the other. What must be the angle between the pass axes of sheets if the intensity of transmitted light is (i) 1/3 of maximum intensity of the transmitted beam and (ii) 1/3 of intensity of incident beam? Assume that a polarizer sheet reduces the intensity of unpolarized light by exactly 50%.

(b) A particle executing SHM along a straight line has velocity 16cm/s and 12cm/s when passing through points 3cm and 4cm from the mean position, respectively. Find (a) amplitude and (ii) period of oscillation.

(c) Explain De Broglie Hypothesis. Compute the De Broglie wavelength of an electron whose kinetic energy is 10 eV.

Use the following:
Planck's Constant, $h = 6.626 \times 10^{-34} m^2 kg/s$,
Speed of light $c = 3 \times 10^8 m/s$
Mass of electron, $m_e = 9.11 \times 10^{-31}$ kg
Elementary Charge, $e = 1.62 \times 10^{-19}$ C

(d) A step index fiber has a core refractive index 1.55 and a cladding of 1.53. If the signal is launched from a medium of refractive index 1.3 Calculate (i) Numerical Aperture (ii) Acceptance angle of the fiber.

UNIT-I

Q2 (a) A monochromatic light beam is incident on the surface of a thin film of thickness t and refractive index μ. With the help of a ray diagram, derive the conditions for intensity maxima and minima in the interference of the reflected beams. (5,5)

(b) A monochromatic light of wavelength $589nm$ is incident normally on a soap film of refractive index 1.42. What is the least thickness of the film that will appear (i) bright and (ii) dark by reflection?

Q3 (a) Derive the intensity distribution function corresponding to diffraction through N slits. Draw the intensity pattern and obtain the conditions for intensity maxima and minima. (5,5)

(b) A grating having 15000 lines per inch produces spectra of a mercury arc. The green line of the mercury spectrum has a wavelength of $546nm$. What is the angular separation between the first order and second order green line?

Q4 **(a)** Solve the differential equation of a damped harmonic oscillator, which is driven by an external sinusoidal force and discuss its solution. (7,3)

(b) Give the condition of Resonance and find an expression for phase and amplitude of the oscillator at resonance.

Q5 **(a)** For a central force, given by $F = -\frac{c}{r^2}\hat{r}$, show that both, the orbital angular momentum L and Areal velocity $\frac{dA}{dt}$ are constant. (5,5)

(b) For a system of two particles interacting under a central force $f(r)\hat{r}$, reduce the two-body problem to equivalent one-body problem. Also, define reduced mass.

UNIT-III

Q6 **(a)** The wave representation of a moving body corresponds to a wave packet. Draw a wave-packet and explain the associated group velocity (v_g) and phase velocity (v_p). (3,7)

(b) Show that the De-Broglie waves associated with a moving particle of mass m and velocity v, travel with group velocity $v_g = v$.

Q7 Solve the time independent Schrodinger Equation to obtain the normalized wave-function ψ_n for a particle confined in a one-dimensional box. Draw the wave-function ψ_n and $|\psi_n|^2$ for $n = 1, 2, 3$. (10)

UNIT-IV

Q8 **(a)** What is the difference between Spontaneous and Stimulated Emission of radiation? (3,7)

(b) Derive Einstein's A and B coefficients.

Q9 Explain the construction and working of an optical fiber-based communication channel with a schematic diagram. (10)